

Abdelrahman Taeha

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Education

Georgia Institute of Technology

M.S. Computer Science (Machine Learning Concentration)

Atlanta, GA
Aug. 2026 – May 2028 (Expected)

California State University, Sacramento

B.S. Computer Science

Sacramento, CA
Aug. 2023 – Dec. 2025

Professional Experience

AI Research Engineer | Medical AI Research

August 2025 – Present

- Co-authored SpineFairBench, a counterfactual vision-language model (VLM) benchmark auditing demographic bias in spinal radiology report generation.
 - Evaluated 7,996 paired radiographs across a frozen 9-model panel with radiologist validation, surfacing demographic recommendation drift across all evaluated models.
 - Developed a neutral baseline model using PyTorch and built data collection pipelines across 2 major open-access clinical repositories to isolate demographic bias in VLM report generation.
- Owned end-to-end development of MedInsider, a FHIR-style benchmark measuring whether medical LLM agents preserve decision integrity when exposed to institutional and financial pressure.
 - Applied feature engineering and model evaluation best practices to construct 840 paired-twin scenarios across 5 regulatory failure families and 8 incentive conditions, isolating when and why agents deviate from compliant behavior.
 - Engineered simulated EHR tooling with deterministic action-log scoring, run auditing, and reproducibility manifests to evaluate tool-use agents across 9 medical vision-language models.

Software Engineer | DHCS

May 2025 – Present

- Automated monitoring and reporting pipelines across enterprise systems, reducing manual observability overhead by 40% and cutting mean time to detection (MTTD) for production incidents in half.
 - Transformed backend integrations and validation workflows that surfaced anomalies early, accelerating root-cause analysis by 35%.
- Established internal dashboards visualizing agent-driven operational metrics, reducing incident resolution time by 30%; improved real-time alerting accuracy by 20% through log-processing pipelines.

Software Engineering Intern (Machine Learning) | District Hut

May 2024 – July 2025

- Scaled distributed data pipelines processing 3 TB/week across cloud infrastructure, improving throughput and system reliability in a high-volume production environment.
 - Revamped SQL data-integrity checks, preventing 5 critical production data-corruption issues and automating 75% of validation tasks with 100% test coverage.
- Collaborated with 3 engineers to automate model retraining agents, reducing collective manual effort by 6 hours per week via autonomous performance metrics.

Project Highlights

APEX — Arrhythmia Pattern Explainer — github.com/abdeltachass/apex-arrhythmia-explainer

- Built an end-to-end ECG decision-support system classifying 71 diagnostic categories, hitting 0.920 macro-AUROC on PTB-XL's held-out test split — matching published benchmarks with a compact 1D CNN; served via FastAPI at 6 ms inference.
- Engineered a reliability layer that flags the model's own untrustworthy output: per-lead Grad-CAM grounding plus four consistency and confidence checks, documented in a Hugging Face model card with a measured sex-performance disparity (0.925 vs 0.906 AUROC, CI excludes zero).
- Reduced Expected Calibration Error from 0.079 to 0.002 (97%) via per-label vector scaling with AUROC unchanged, cutting spurious findings 93% (5.09 → 0.35 per record); diagnosed why standard temperature scaling failed.

SENTRY — Vision-Language Incident Report Generator — github.com/abdeltachass/SENTRY

- Designed and deployed an end-to-end computer vision and NLP system generating security-incident reports from camera frames, pairing a CLIP-style image encoder with a LoRA-fine-tuned language model to adapt a pretrained model efficiently.
- Quantified vision-language model hallucination rate against 10,000 manually reviewed outputs, reducing ungrounded claims by 35% and improving overall output reliability.

S.A.I.N.T. — Network Intrusion Detection System — github.com/abdeltachass/S.A.I.N.T.

- Partnered with 2 peers to build an anomaly detection reasoning engine for network threats, reaching 89% accuracy on NSL-KDD (125K+ samples) via feature engineering across 41 traffic-flow features.
- Designed a low-latency inference agent using Flask + Redis + Plotly, processing 50–100 network connections/sec with an interactive agent-in-the-loop dashboard for analyst review.

Skills

Languages: Python, Java, C/C++, SQL, JavaScript, Swift, MATLAB, R

ML & AI: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Model Evaluation

AI & Agents: Autonomous Agents, LLM Orchestration (LangChain/CrewAI), RAG, Prompt Engineering

Frameworks & Cloud: React.js, Node.js, Spring Boot, Flask, FastAPI, AWS, GCP, Docker, Git, GitHub Actions, CI/CD

Research & Domains: VLM Evaluation, Counterfactual Benchmarking, Medical Imaging, Demographic Bias Auditing, SIEM, Monitoring Pipelines